

Lab 1: Requirements Traceability Matrix (RTM)

Problem Statement #23: Hyperlocal Courier Dispatch & Tracking Engine

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Req ID	Requirement	Priority	Use Case(s)	Epic	Kanban Story	Scrum Story	Bug (Failure Condition)
FR-001	Match a courier request with the nearest active rider within 3 km	High	UC-02 Match Nearest Active Rider (included by UC-01)	Courier Request and Dispatch	KBPS25-11	SBPS23-11	BTBPS23-1 Offline or farther rider assigned
FR-002	Sender creates a courier request with pickup, delivery and parcel details	High	UC-01 Create Courier Request	Courier Request and Dispatch	KBPS25-12	SBPS23-12	BTBPS23-2 Request created with required details missing
FR-003	Rider accepts or rejects an assigned courier request	High	UC-03 Receive / View Dispatch	Courier Request and Dispatch	KBPS25-13	SBPS23-13	BTBPS23-3 Rejected request stays assigned to same rider
FR-004	Generate a delivery route and optimize the order of stops	High	UC-06 Plan Delivery Route UC-07 Optimize Multi-Stop Route	Routing and Live Tracking	KBPS25-14	SBPS23-14	BTBPS23-4 Delivery route is missing assigned stops
FR-005	Verify an OTP at the destination before marking the courier delivered	High	UC-08 Complete Delivery UC-09 Verify Destination OTP	Delivery Completion and Reliability	KBPS25-15	SBPS23-15	BTBPS23-5 Incorrect OTP allows delivery completion
NFR-001	Status and live GPS updates reach the Sender Client in under 2 seconds	High	UC-04 Track Delivery UC-05 Update Delivery Status	Routing and Live Tracking	KBPS25-16	SBPS23-16	BTBPS23-6 GPS and status updates exceed 2 seconds
NFR-002	System availability of at least 99.5% during normal operation	Medium	All use cases (system-wide)	Delivery Completion and Reliability	KBPS25-17	SBPS23-17	BTBPS23-7 System availability falls below 99.5%

Traceability key

Use Case(s): from the UML use-case diagram (UM_diagram.pdf). Epic: Jira epic the requirement is grouped under.

Kanban Story: Kanban_BPS#23 project (KBPS25). Scrum Story: Scrum_BPS#23 project (SBPS23). Bug: Bug_Tracking_BPS#23 project (BTBPS23), derived from each requirement's Fail condition.